Majid Ghasemi

Ph.D. Student, Electrical & Computer Engineering, University of Waterloo

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Research Interests

Social and multi-agent reinforcement learning, machine ethics, and AI alignment. My work asks how ethical behaviour can be learned and transmitted between agents rather than hard-coded as a fixed reward, contributing to safe AI.

Education

University of Waterloo — Waterloo, ON, Canada
Ph.D., Electrical & Computer Engineering — *Sep 2025 – Present*
Advisor: Mark Crowley
Focus: Ethical Decision Making in Reinforcement Learning

Wilfrid Laurier University — Waterloo, ON, Canada
M.A.Sc., Applied Computing (GPA A+) — *Sep 2023 – May 2025*
Advisor: Dariush Ebrahimi
Thesis: *Reinforcement Learning for Scheduling and Routing: Electric Vehicle Charging and Patrol Scheduling*

Imam Khomeini International University — Qazvin, Iran
B.Sc., Software Engineering (GPA A–) — *Sep 2017 – Mar 2022*
Advisor: Mahdi Bahaghighat Thesis: Machine Learning for Phishing Websites Detection

Publications

Peer-Reviewed Conference Papers

[C1] **The Forager's Dilemma: Peer Enforcement Alone Is Not Enough for Stable Multi-Agent Alignment.**
Majid Ghasemi, Prabhjyot Singh, Mark Crowley. *Ninth AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, 2026.

[C2] **AI Alignment without Representation: From Scalar Rewards to Social Deliberation.**
Majid Ghasemi, Mark Crowley. *Ninth AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, 2026.

[C3] **Scaling Limits of Deep Reinforcement Learning: A Stability Analysis with Maximal Update Parametrization.**
Majid Ghasemi, Mark Crowley. *The 39th Canadian Conference on Artificial Intelligence*, 2026.

[C4] **Adaptive and AI-Driven Vehicle Patrol Scheduling with Integrated Emergency Response.**
Majid Ghasemi, Dariush Ebrahimi. *IECON 2025, 51st Annual Conference of the IEEE Industrial Electronics Society*, 2025.

[C5] **Optimal and Heuristic Solutions for Efficient Vehicle Patrol Scheduling in Dynamic Urban Safety.**
Majid Ghasemi, Ebrahim Sarkhouh, Fadi Alzhouri, Dariush Ebrahimi. *IEEE 34th International Symposium on Industrial Electronics (ISIE)*, 2025.

[C6] **Comparative Analysis of Reinforcement Learning for Reliable Real-Time EV Routing and Charging.**
Majid Ghasemi, Fadi Alzhouri, Dariush Ebrahimi. *21st International Conference on the Design of Reliable Communication Networks (DRCN)*, 2025.

[C7] **Navigating Smart Order Routing in Fixed Income Trading with Linear Programming and Dijkstra's Algorithm.**
Cassel Robson, Marko Misic, Erika Capper, Mila Cvetanovska, Majid Ghasemi, Fadi Alzhouri, Dariush Ebrahimi. *International Conference on the Dynamics of Information Systems*, 2024.

Journal Articles

[J1] **Efficient Vehicle Patrol Scheduling for Urban Safety: Optimization and Heuristic Approaches.**
Majid Ghasemi, Ibrahim Sorkhoh, Fadi Alzhouri, Dariush Ebrahimi. *Future Generation Computer Systems*, 2026.

[J2] **Reinforcement Learning and Model-based Planning in Practice: A Survey of Algorithmic Rationale and Domain Applications.**
Majid Ghasemi, Amirhossein Mousavi, Dariush Ebrahimi. *Artificial Intelligence Review*, 2026.

[J3] **Real-Time Electric Vehicle Routing and Charging Optimization with Battery Life Consideration.**
Dariush Ebrahimi, Majid Ghasemi, Mohammad Kashefi, Fadi Alzhouri. *IEEE Internet of Things Journal*, 2026.

[J4] **Citation Count Prediction Based on Google Scholar Profiles and Clarivate's Journal Citation Reports.**
Mahdi Bahaghighat, Leila Akbari, Majid Ghasemi, Qin Xin. *Information Research*, 2023.

[J5] **A High-Accuracy Phishing Website Detection Method Based on Machine Learning.**
Mahdi Bahaghighat, Majid Ghasemi, Figen Ozen. *Journal of Information Security and Applications*, vol. 77, 2023.

Peer-Reviewed Workshop Papers

[W1] **Learning When to Trust in Contextual Social Bandits.**
Majid Ghasemi, Mark Crowley. *19th European Workshop on Reinforcement Learning (EWRL)*, 2026.

[W2] **Rethinking AI Alignment: From Static Rewards to Social Reinforcement Learning.**
Majid Ghasemi, Mark Crowley. *Pluralistic Alignment Workshop, ICML*, 2026.

[W3] **Can Standard MARL Metrics Distinguish Communicative from Strategic Action?**
Majid Ghasemi, Mark Crowley. *PhilML Workshop, ICML*, 2026.

[W4] **Toward Virtuous Reinforcement Learning: A Critique and Roadmap.**
Majid Ghasemi, Mark Crowley. *Machine Ethics Workshop (From Formal Methods to Emergent Machine Ethics), AAAI*, 2026.

Preprints and Technical Reports

[P1] **Objective Decoupling in Social Reinforcement Learning: Recovering Ground Truth from Sycophantic Majorities.**
Majid Ghasemi, Mark Crowley. *arXiv preprint*, 2026.

[P2] **Comprehensive Survey of Reinforcement Learning: From Algorithms to Practical Chal-**

lenges.
Majid Ghasemi, Amirhossein Mousavi, Dariush Ebrahimi. *arXiv preprint*, 2024.

[P3] **Introduction to Reinforcement Learning.**
Majid Ghasemi, Dariush Ebrahimi. *arXiv preprint*, 2024.

Industry Experience

Kisoji Biotechnology Inc. Waterloo, ON, Canada
AI Research Intern, Machine Learning Department *May 2026 – August 2026*

- Building the Expressibility Engine, a three-module system for antibody expression optimization.
- Domain-adaptive prediction with Bayesian project embeddings; RL-guided experimental design using GRPO with DPP-based diversity; sparse graph-based causal driver discovery.
- Reduces wet-lab iteration through active learning while producing interpretable guidance for antibody engineers.

Honours and Awards

AIES Student Program Scholarship Award	*2026*
Graduate Student Research and Development Award (×2), University of Waterloo	*2026*
International Doctoral Student Award, University of Waterloo	*2025 – 2029*
Academic Excellence Gold Medal, Wilfrid Laurier University	*2025*
Graduate Scholarship, Wilfrid Laurier University	*2023 – 2025*
Graduate Fellowship, Wilfrid Laurier University	*2023 – 2025*
Differential Tuition Fee Waiver, Wilfrid Laurier University	*2023 – 2025*
Excellence Students Tuition Fee Waiver, Imam Khomeini Intl. University	*2017 – 2022*

Grants

HU University of Applied Sciences Utrecht Aug 2022 – Aug 2023
Research grant to study the Woman, Life, Freedom movement in Iran using computational social science and machine learning methods.

Research Experience

UWECEML / CompThink Lab, University of Waterloo Sep 2025 – Present
Graduate Researcher, advised by Mark Crowley

- Developing socially informed and ethically grounded methods for multi-agent reinforcement learning.
- Work spans virtue transmission in sequential social dilemmas, contextual bandits with strategic feedback, and constrained ethical agent design.

Edge Computing Lab, Wilfrid Laurier University Sep 2023 – May 2025
Graduate Researcher, advised by Dariush Ebrahimi

- Reinforcement learning for intelligent transportation, focusing on EV fleet management, routing, and charging optimization.
- Developed exact and heuristic baselines for urban patrol scheduling under dynamic demand.

AIST Lab, Imam Khomeini International University Jan 2021 – Mar 2022

Undergraduate Researcher, advised by Mahdi Bahaghighat

- Machine learning methods for phishing website detection.

Invited Talks and Presentations

Designing Virtuous Agents with Reinforcement Learning, Formal Ethics 2026, Buffalo, USA *Jul 2026*
Optimal and Heuristic Solutions for Vehicle Patrol Scheduling, IEEE ISIE 2025 *Jun 2025*

Teaching Experience

Teaching Assistant 2021 – Present

- **University of Waterloo:** ECE108 Discrete Mathematics and Logic (S26), ECE406 Algorithm Design and Analysis (W26), ECE250 Data Structures and Algorithms, tutor (F25).
- **Wilfrid Laurier University:** CP468 Artificial Intelligence (F23, F24, W25), CP421 Data Mining (W24), CP312 Algorithm Design and Analysis (W24).
- **Imam Khomeini Intl. University:** Artificial Intelligence, Data Mining, Fundamentals of Programming (2021 – 2022).
- Guest lectured selected topics, authored quizzes and project specifications, designed and evaluated term projects, and held regular office hours.

Academic Service

Program Committee

- AAAI/ACM Conference on AI, Ethics, and Society (AIES), 2026

Reviewer

- Reinforcement Learning in Big Worlds Workshop, Reinforcement Learning Conference (RLC), 2026
- PhilML Workshop, ICML 2026
- Pluralistic Alignment Workshop, ICML 2026
- Post-AGI Science and Society (P-AGI) Workshop, ICLR 2026
- IEEE Canadian Journal of Electrical & Computer Engineering (CJECE)
- ACM Transactions on Intelligent Systems and Technology (TIST)

Technical Skills

Languages: Python, C++, JavaScript, HTML/CSS
ML frameworks: PyTorch, JAX, TensorFlow
Scientific stack: NumPy, Pandas, Matplotlib
Spoken languages: English (fluent), Persian (native), Azeri (native)